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CGIAR SCALING FOR IMPACT

Section 0 – Title and contact Information

- Title of the proposed solution track: Scaling Forage-Based Livestock Feed Innovations for Resilient Rural Livelihoods in Ethiopia and Zambia
- Title of the AoW2 Flagship that the solution track aligns with: Flagship 2.3: Delivering Sustainable Animal and Aquaculture Genetics, Inputs, and Services
- Lead Scientist(s) / Contact Person(s) / Center(s) : Million Gebreyes, Kindu Mekonnen, Melkamu Derseh, BALENI, Thembinkosi, Getachew Feye/ International Livestock Research Institute
- Emails of the Lead Scientist(s) / Contact Person(s) / Center(s): m.gebreyes@cgiar.org / t.baleni@cgiar.org
- Key CGIAR Center Partners (in the case of multi-Center EoIs): ILRI/ With a possibility of working with ABC (in Ethiopia) and CIMMYT (in Zambia)
- Technical link to other Science Programs – Sustainable Animal and Aquatic Food Systems and Sustainable Farming Program

Section 1 – Innovation Package, Value Proposition & Alignment

- **The problem to be solved:** Ethiopia and Zambia face critical challenges in livestock productivity due to inadequate feed systems. In Ethiopia, despite having one of Africa's largest livestock populations, productivity remains low—primarily due to poor feed quality and availability. Smallholders rely on declining natural pastures and crop residues, while commercial feeds are unaffordable. In Zambia, high feed costs, climate-induced livestock losses, and limited market access threaten rural incomes and resilience. The proposed Scaling Solution aims to commercialise and scale forage-based feed innovations through digitally enabled bundles, policy support, and inclusive business models. It targets smallholder farmers, with a strong focus on empowering women and youth through forage seed and biomass enterprises, SME development, and rural industrialisation. By addressing feed shortages, climate risks, and economic exclusion, the solution enhances resilience and unlocks sustainable livelihood opportunities.
- **Your scaling ambition:** By 2028, our integrated initiative will establish robust, inclusive feed innovation systems in Ethiopia and Zambia. In Ethiopia, we will scale up cultivated forage use to 250,000 farmers, creating/improving 10,000 farm jobs, doubling average milk yields, and improving household incomes, with a strong focus on gender and youth inclusion. This will shift feed systems towards sustainable, forage-based models, enhancing climate resilience and creating new business opportunities. In Zambia, mass communication and market systems development will reach over 500,000 individuals, train 400 extension personnel, and create 10,000 jobs through new rural feed enterprises. Farmers will be empowered to produce and home-mix nutritionally complete ruminant diets, improving productivity and resilience. Collectively, these efforts will accelerate progress toward 2030 targets—expanding sustainable feed solutions to millions, generating thousands of jobs, improving nutrition, and restoring productive land—driving inclusive, climate-resilient growth in Africa's livestock sector.
- **Innovation & Value Proposition** – Ethiopia's Digitally Enabled Forage-Based Feed Innovation Bundle is a mature, scalable solution integrating national-level data pooling, standardised analytics, and advisory systems. Years of validation and piloting underpin its novelty, with feed balance mapping and tailored utilisation strategies optimising resource

use and boosting productivity across dairy, poultry, and feedlot systems. Market-led seed system development strengthens both formal and informal networks, while scaling pathways promote commercialisation through behavioural change and policy support. In Zambia, smallholder farmers are empowered to cultivate and home-mix nutritionally complete ruminant diets using mucuna and lablab, proven competitive with commercial feeds. These innovations deliver measurable gains in productivity, resilience, and nutrition, while reducing risk and strengthening institutional capacity for ministries, agribusinesses, and end-users. Both approaches are supported by strong partnerships and align with science programmes such as SAAFS, Sustainable Farming, and Multifunctional Landscapes, offering synergistic opportunities for scaling impact across Africa's livestock sector.

- **Customers & Alignment** – In Ethiopia, main institutional customers include ministries, donor programmes (e.g. World Bank, IFAD), forage and forage seed agribusinesses, and lenders supporting flagship initiatives like the Livestock and Fishery Sector Development Program and the Lowland Resilience Program. End-users are farmers, SMEs, cooperatives, and extension systems. In Zambia, key partners span microfinance institutions, agricultural retailers, public extension services, research organisations, machinery suppliers, and forage seed producers. End-users include smallholder farmers, agripreneurs, and local councils. Both tracks align with AoW2 Flagship 2.3 and Grand Challenge 2, delivering proven feed innovations and advisory services to drive inclusive, market-led scaling and sustainable livestock development.
- **Evidence of Impact** – ILRI's forage innovations in Ethiopia have demonstrated measurable impact on livestock productivity and farmer livelihoods. Pilots under Africa RISING, Mixed Farming Systems, and SAPLING validated niche-compatible forage species and seed systems, with adoption driven by bundled advisory services, behavioural change communication, and policy support (https://doi.org/10.1002/agj2.20853). Early results from forage commercialisation show increased productivity and farmer incomes (https://hdl.handle.net/10568/170114), with national partners and donor-backed projects (AICCRA, TAAT) confirming scalability. In Zambia, ILRI research shows beef pen-feeding trials increased net profit by 7–10% versus commercial supplements (https://hdl.handle.net/20.500.11766/4940), and mucuna/lablab-based rations raised dairy farmer incomes by 16.5–17%. Pen-feeding with home-grown feeds boosted household income and livestock value, with average gross income rising to $650/year and market value of beef animals doubling in 60 days. These results confirm the effectiveness and scalability of the innovations.

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

- **Scaling Pathway & Sustainability** – In Ethiopia, the scaling pathway is evolving from donor dependence to a market-driven, sustainable model. Alongside public-private partnerships (PPP), we are actively engaging private businesses, notably the Ethiopian Forage Seed Production and Marketing Association and private forage seed enterprises, as primary outlets for forage seed distribution. Together with SNV, we are developing market-based seed delivery mechanisms. Farmer cooperatives are being supported to produce and market forage seeds and biomass, with targeted business development support to enhance their reach and viability. This approach ensures that, as catalytic funding phases out, farmer-led enterprises, private sector actors, and cooperatives will maintain and expand the system through cost recovery, commercial partnerships, and gradual transition to self-sustaining market systems. Sustainability is further reinforced through training, extension support, and the promotion of climate-resilient forage varieties.

In Zambia, the scaling approach is structured around two distinct pathways. The private sector-led pathway focuses on commercialisation, mechanisation, and financial access, supporting rural feed enterprises and SMEs through activities such as market facilitation, business incubation, and partnerships with microfinance institutions, machinery suppliers, and seed producers. The public sector-supported pathway leverages digital extension, mass media, and social media campaigns to build awareness and capacity, aiming to reach 500,000 farmers. Key activities under this pathway include training 400 extension officers, engaging community influencers, and implementing CRM systems, bulk messaging, and outgrower schemes to ensure widespread adoption and sustainability. By clearly distinguishing these pathways and their associated activities, the approach ensures systematic, inclusive, and sustainable scaling, with each partner playing a defined role in strengthening the value chain and ensuring long-term viability at scale.

- **Partnerships & Leverage** – In Ethiopia, the Digitally Enabled Forage-Based Feed Innovation Bundle is championed internally by ILRI's multidisciplinary team, including crop-livestock systems scientists, animal nutritionists, agricultural economists, and scaling specialists. The track is technically linked to CGIAR science programmes such as Sustainable Animal and Aquatic Foods and Sustainable Farming. A central feature is the strong partnership with private sector actors, notably the Ethiopian Forage Seed Production and Marketing Association (EFSPMA) and private forage seed businesses. ILRI provides technical capacity building to EFSPMA members—supporting seed quality assurance, business development, and market intelligence—while the association leads a market-based approach for forage seed scaling, serving as the main outlet for seed distribution and commercialisation. Collaboration with SNV is developing market-based seed delivery mechanisms, and farmer cooperatives are being enabled to produce and sell forage seeds and biomass, diversifying scaling pathways. Evidence of demand includes formal collaboration requests from national partners, alignment with Ethiopia's Food Systems Transformation Pathway, and integration into the National Dairy Innovation and Development Platform. Budget allocations from ILRI and other CGIAR centres are mobilised through AICCRA and TAAT. Major non-CGIAR programmes leveraged include the Livestock and Fishery Sector Development Programme and the Lowland Resilience Programme (IFAD, World Bank), with additional prospects from the African Development Bank, AU-IBAR, the Gates Foundation, AGRA, and the Mastercard Foundation.

  In Zambia, the partnership ecosystem includes IMWI (water management and CAPEX finance), CIMMYT (agronomic expertise), and NARES (on-farm testing, policy, and market research). Other partners are Msekera Research Station, Nico Insurance Company (insurance/loans), INDO Bank Zambia (agriculture finance), Modern Bazaar (retail/wholesale), and K2 (forage seed production). Evidence of demand is seen in collaboration requests, policy alignment, and budget allocations, with commitment shown through MOUs, LOIs, and co-investment. The track leverages government livestock programmes, donor-funded market systems initiatives, and private sector investments for sustainable, large-scale impact.

- **Risks & Bottlenecks** – In Ethiopia, scaling a market-based forage seed system faces several context-specific risks and bottlenecks. The most significant policy risk is the entrenched practice of indiscriminate free distribution of forage seeds, which undermines market development and discourages private sector investment. To mitigate this, we will implement a structured policy dialogue with the Ministry of Agriculture, regional bureaus, and key stakeholders to develop clear guidelines on when free seed delivery is appropriate (e.g., emergencies) and when to prioritise market-based delivery. This will be supported by milestones such as forming a policy working group, drafting new policy instruments, and securing official endorsement. Operationally, the bulkiness of forage biomass creates logistical challenges, as high transport costs can deter private sector engagement. We will promote cost-effective densification techniques—such as baling and compacting—to make transport more feasible for small-scale suppliers. Security risks from local conflicts may disrupt implementation; careful site selection and strong local partnerships will help ensure continuity. Social acceptance is another challenge, especially in transitioning from free to paid seed systems. We will engage farmer groups and cooperatives through participatory approaches and targeted awareness campaigns, highlighting the long-term benefits of sustainable, market-based seed systems and building trust in commercial seed sources. In Zambia, knowledge gaps in livestock extension and research systems are a bottleneck. S4I will prioritise capacity building for extension personnel and engage private extension organisations for training-of-trainers programmes. Regulatory and policy barriers for rural SMEs will be addressed by IMWI through AoW3, supporting policy and regulatory easing to create an enabling environment for innovation and sustainable scaling.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

- **Readiness & Responsible Scaling** – In Ethiopia, the market-based forage seed system demonstrates strong readiness for scaling, having achieved a scaling readiness score of 6 in 2022 (see https://reporting.cgiar.org/reports/result-details/1886?phase=1). Innovation packaging of a commercial business model for forage and forage seed was done in 2023 (https://reporting.cgiar.org/reports/ipsr-details/4943?phase=2). The next step is to develop a scaling strategy that integrates responsible scaling principles, with a focus on youth employment and significant opportunities for female youth in rural areas. This inclusive approach will strengthen local ownership and sustainability. Environmental safeguards will be embedded through the promotion of climate-resilient forage species and sustainable land management. The pathway will be co-developed with national and local partners to ensure alignment with policy frameworks and social equity.

  In Zambia, the innovation has been profiled and packaged, achieving a scaling readiness and use level of 8. Responsible scaling is ensured by targeting women, youth, and marginalised groups through capacity building, inclusive business models, and environmentally sustainable practices, supporting broad-based adoption and long-term impact. Through 2025, ILRI has managed to identify and mirror key partners identified in the scaling report.

  Achievable target, Resources & Activities – For Ethiopia, from 2026 to 2028, the programme aims to reach 100,000 smallholder farmers across four regions, each cultivating at least one or two forage crops to boost livestock productivity and build sustainable feed systems. This will translate into reaching 250,000 by 2030. The first year will focus on launching region-specific scaling strategies, followed by rapid expansion in year two, and amplification and replication in year three. The initiative will engage 40 farmer organisations (10 per region), each with around 1000 members, and actively involve youth groups and women's associations to ensure inclusivity. For Zambia, over the same period, advisory systems will reach 500,000 farmers through radio, TV, mass messaging, fairs, traditional leaders, and extension systems, aiming 90% of adoption of leguminous forage based feed ration. Land usage will expand to 100,000 hectares (80% adoption), with 10,000 tonnes of fodder and seed produced and 40,000 livestock fed. Financial leverage targets $1 million through loans, feed sales, and bilateral projects. Activities include outgrower schemes, market linkages, pen fattening, and capacity building for extension and farmer groups, ensuring sustainable adoption and impact.

Key resources required include a dedicated team: a Technical Coordinator (forage agronomy), a Scaling Coordinator (partnerships and process), and a Partnership Manager (policy, advisory, market development). Institutional support will come from ILRI, national research systems, and regional bureaus. Financial resources will be mobilised through S4I, donor projects, and public-private partnerships.

Core activities include training and capacity development, policy dialogue, extension message development, innovation packaging and scaling strategy development, business readiness support for farmer groups and youth-led enterprises, and the rollout of market-based seed and forage delivery systems.

Timeline, Milestones & Monitoring – From 2026 to 2028, the programme will reach 100,000 smallholder farmers in Ethiopia and 200,000 in Zambia, scaling up to 250,000 and 500,000 respectively by 2030. Annual milestones include: (i) expanding farmer participation in forage cultivation, (ii) forming and strengthening farmer organisations and youth-led enterprises, (iii) training 1,000 extension personnel in Ethiopia and 400 in Zambia, (iv) adoption of supportive policies through ongoing policy dialogue, and (v) business development for 100 farmer businesses across both countries. By 2028, 80% of targeted Zambian farmers will be producing leguminous forage crops, with 10 rural cooperatives established and robust partnerships with seed houses, insurance, and microfinance institutions.

Progress will be monitored using digital Customer Relation Management systems, extension reports, and regular partner reviews. Key indicators—such as user reach, adoption rates, partnerships formed, business growth, and policy changes—will be tracked quarterly. Lessons will be captured through MELIA studies, impact assessments, and stakeholder feedback sessions. These insights will inform adaptive management, allowing the team to refine strategies, address bottlenecks, and ensure inclusivity. Annual reflection workshops will be held to review progress, share learning, and co-design adaptations with partners and beneficiaries, ensuring the scaling track remains responsive and impactful.

Additional Annexes

- IPSR Innovation Profile
  - Ethiopia: https://reporting.cgiar.org/reports/result-details/1886?phase=3
  - Zambia: https://reporting.cgiar.org/reports/result-details/782?phase=5
- Scaling Readiness assessment results
  - Ethiopia: https://reporting.cgiar.org/reports/ipsr-details/4943?phase=5
  - Zambia: https://reporting.cgiar.org/reports/ipsr-details/7949?phase=5
- MELIA studies, impact assessments, reports, publications

Ethiopia

- https://hdl.handle.net/10568/163127
- https://doi.org/10.1002/agj2.20853
- https://hdl.handle.net/10568/134733
- https://hdl.handle.net/10568/170114
- https://hdl.handle.net/10568/163127
- https://reporting.cgiar.org/reports/result-details/12546?phase=4
- https://reporting.cgiar.org/reports/result-details/18510?phase=4
- https://reporting.cgiar.org/reports/result-details/1718?phase=1
- https://reporting.cgiar.org/reports/result-details/18533?phase=4
- https://reporting.cgiar.org/reports/result-details/12753?phase=4
- https://reporting.cgiar.org/reports/result-details/12480?phase=4
- https://reporting.cgiar.org/reports/result-details/6942?phase=3

Zambia

- https://hdl.handle.net/10568/174947
- https://hdl.handle.net/20.500.11766/5638
- https://hdl.handle.net/20.500.11766/4940
- https://hdl.handle.net/10568/120615
- Forage-based livestock fattening rations gaining popularity with smallholder beef and goat producers in Zimbabwe - CGIAR